Marco Del Fava

Summary

Recently admitted to the UC Davis M.S. in Biomedical Engineering (Fall 2026) and active neuroscience researcher working at the intersection of human intracranial electrophysiology, single-unit digital signal processing, and neurotechnology tooling. Experience includes spike sorting with Kilosort4, O-Sort, and SpikeInterface; building GPU-accelerated pipelines on AWS; and implementing Dark Horse Neuro / Neuralynx acquisition workflows for replay, compression-aware MED recording, and large-scale neural data handling. I look forward to a career in neural signal processing, neural acquisition systems, and BCI-related infrastructure.

GitHub https://github.com/BarkBarkBarkBarkBarkBarkBark

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Selected Technical Projects

Dark Horse Neuro / Neuralynx acquisition pipeline — 2026

- Built a Python orchestration layer to launch DHN-AQ headlessly over SSH, generate RC/CS session files, and coordinate Pegasus / NetCom replay workflows for human intracranial recordings.
- Implemented an NRD replay bridge that re-broadcasts ATLAS-format UDP packets so archived Neuralynx sessions can be ingested by DHN-AQ as if hardware were live.
- Automated compression-aware MED recording setup for 512-channel macro / micro configurations, including decimation, lossless/lossy options, and pre-flight network diagnostics.

spike_discrim single-unit DSP workbench — 2026

- Benchmarked low-latency waveform representations for real-time single-unit discrimination using curated OSort snippets, synthetic data, AUC, balanced accuracy, mutual information, and KNN purity.
- Built a Numba-accelerated WeightBank population code to score spike-likeness and project waveforms into afferent activation space for clustering and unit-separation analysis.

SNN Agent real-time spike sorting — 2026

- Developed a streaming spiking neural network with bandpass filtering, Kalman noise gating, competitive STDP, and browser-based monitoring for live spike detection and sorting.
- Evaluated 20 kHz real-time pipeline behavior across synthetic, .ncs, UDP, and LSL inputs with online decoding for closed-loop neurotech use cases.

Education

M.S. Biomedical Engineering, University of California, Davis — Davis, CA — Beginning Fall 2026 Recently admitted; incoming first-year master's student

Python for Data Science, Professional Concentration, UC Davis Continuing and Professional Education — Davis, CA — 2025

M.S. Psychology, Arizona State University — Tempe, AZ — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — Davis, CA — 2017

Professional Experience

Operations Specialist, First Step Communities — Sacramento, CA Jul 2021 — Present

- Analyzed program and partner datasets using Python, SPSS, and Excel to identify predictors of successful exits and service effectiveness.
- Coordinated data collection from clinical partners including UC Davis Health and Kaiser Permanente to support integrated care outcome analyses.
- Identified an association between on-site clinical engagement and permanent housing outcomes; supported leadership in scaling integrated clinical services across sites.
- Developed 40+ standardized data collection forms and automated reporting workflows using Microsoft Power Automate.

Analyst, Eurofins Test America — West Sacramento, CA Oct 2019 — Jun 2021

- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.
- Performed solvent extraction of organic compounds from soil and water samples in a high-throughput production setting.
- Coordinated daily scheduling and backlog management to meet turnaround targets across teams.

Skills & Training

Neural Data & Signal Processing:

- Single-unit DSP and waveform feature extraction
- Spike sorting (Kilosort4, O-Sort, SpikeInterface)
- Neural data processing (filtering, PCA, template matching, clustering)
- Intracranial electrophysiology (microwire, LFP, iEEG)

Neural Acquisition & Tooling:

- Neuralynx Pegasus / ATLAS / NetCom workflows
- Dark Horse Neuro DHN-AQ RC/CS configuration
- NRD replay, UDP relay, and MED archive generation
- Linux, SSH automation, xvfb, and acquisition diagnostics

Machine Learning & Data Science:

- Python (NumPy, Pandas, scikit-learn, PyTorch, Numba)
- Statistical analysis (SPSS)
- SQL and database engineering (PostgreSQL, PGVector)
- Cloud GPU computing (AWS EC2, CUDA, Docker) and MATLAB workflows

General Technical:

- Git and version control
- Linux and shell scripting

- Web interfaces (Streamlit)
- Microsoft 365 administration

Data Systems & Reporting:

- Dashboards and stakeholder reporting (Looker)
- Workflow automation (Microsoft 365, Power Automate)
- Survey design and focus group facilitation
- Equity-informed outcomes measurement

Certifications

- **Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries** — UC Davis Continuing and Professional Education (2025-09-18) — Badge- **CITI — Biomedical Research** — CITI Program- **CITI — Essentials of Research Administration** — CITI Program- **CITI — GCP Social and Behavioral Research Best Practices** — CITI Program